
PHYSICS

9702/35

Paper 3 Advanced Practical Skills 1

May/June 2019

MARK SCHEME

Maximum Mark: 40

Published

This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination. It shows the basis on which Examiners were instructed to award marks. It does not indicate the details of the discussions that took place at an Examiners' meeting before marking began, which would have considered the acceptability of alternative answers.

Mark schemes should be read in conjunction with the question paper and the Principal Examiner Report for Teachers.

Cambridge International will not enter into discussions about these mark schemes.

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This document consists of 7 printed pages.



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PUBLISHED**Generic Marking Principles**

These general marking principles must be applied by all examiners when marking candidate answers. They should be applied alongside the specific content of the mark scheme or generic level descriptors for a question. Each question paper and mark scheme will also comply with these marking principles.

GENERIC MARKING PRINCIPLE 1:

Marks must be awarded in line with:

- the specific content of the mark scheme or the generic level descriptors for the question
- the specific skills defined in the mark scheme or in the generic level descriptors for the question
- the standard of response required by a candidate as exemplified by the standardisation scripts.

GENERIC MARKING PRINCIPLE 2:

Marks awarded are always **whole marks** (not half marks, or other fractions).

GENERIC MARKING PRINCIPLE 3:

Marks must be awarded **positively**:

- marks are awarded for correct/valid answers, as defined in the mark scheme. However, credit is given for valid answers which go beyond the scope of the syllabus and mark scheme, referring to your Team Leader as appropriate
- marks are awarded when candidates clearly demonstrate what they know and can do
- marks are not deducted for errors
- marks are not deducted for omissions
- answers should only be judged on the quality of spelling, punctuation and grammar when these features are specifically assessed by the question as indicated by the mark scheme. The meaning, however, should be unambiguous.

GENERIC MARKING PRINCIPLE 4:

Rules must be applied consistently e.g. in situations where candidates have not followed instructions or in the application of generic level descriptors.

GENERIC MARKING PRINCIPLE 5:

Marks should be awarded using the full range of marks defined in the mark scheme for the question (however; the use of the full mark range may be limited according to the quality of the candidate responses seen).

GENERIC MARKING PRINCIPLE 6:

Marks awarded are based solely on the requirements as defined in the mark scheme. Marks should not be awarded with grade thresholds or grade descriptors in mind.

Question	Answer	Marks
1(a)(i)	Value of c in the range 55.0–65.0 cm with unit.	1
1(a)(ii)	Value of I in the range 75.0–175.0 mA with unit.	1
1(b)	Six sets of readings of c and I (different values) showing the correct trend and without help from the Supervisor scores 5 marks, five sets scores 4 marks etc.	5
	Range: Includes a reading of $c \geq 95.0$ cm and no values of $c < d$.	1
	Column headings: Each column heading must contain a quantity, a unit and a separating mark where appropriate. There should be no units for the n or $(n + 2) / (n + 1)$ columns. The presentation of the quantity and unit must conform to accepted scientific convention e.g. c / m .	1
	Consistency: All values of c must be given to the nearest mm only.	1
	Calculation: Values of $(n + 2) / (n + 1)$ are correct.	1
1(c)(i)	Axes: Sensible scales must be used, no awkward scales (e.g. 3:10 or fractions). Scales must be chosen so that the plotted points occupy at least half the graph grid in both x and y directions. Scales must be correctly labelled with the quantity that is being plotted. Scale markings should be no more than three large squares apart.	1
	Plotting of points: All observations in the table must be plotted on the grid. Diameter of plotted points must be $\leq$ half a small square (no “blobs”). Points must be plotted to an accuracy of half a small square.	1
	Quality: All observations in the table (at least 4) must be plotted on the grid. Trend of points on graph must be correct. It must be possible to draw a straight line that is within ± 0.05 (to scale) of all the plotted points on the $(n + 2) / (n + 1)$ axis.	1

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Question	Answer	Marks
1(c)(ii)	Line of best fit: Judge by balance of all points on the grid about the candidate's line (at least 5 points). There must be an even distribution of points either side of the line along the full length. If there are 6 or more points, allow one anomalous point only if clearly indicated by the candidate. Lines must not be kinked or thicker than half a small square.	1
1(c)(iii)	Gradient: The hypotenuse of the triangle used must be greater than half the length of the drawn line. The method of calculation must be correct. Do not allow $\Delta x / \Delta y$. Both read-offs must be accurate to half a small square in both the x and y directions. Sign of gradient must match graph.	1
	y-intercept: Correct read-off from a point on the line and substituted into $y = mx + c$. Read-off must be accurate to half a small square in both x and y directions. or Intercept read directly from the graph with read-off at $x = 0$, accurate to half a small square.	1
1(d)	Value of S = candidate's gradient and value of T = candidate's intercept. The values must not be fractions.	1
	Unit for S and T correct (mA or A).	1
1(e)	Correct line W with smaller gradient and below the original line (with a smaller intercept if false origin).	1

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Question	Answer	Marks
2(a)	Raw values of x and L_1 to the nearest mm.	1
	Mean values of x and L_1 in the range 23.0–27.0 cm.	1
2(b)(i)	Values of L_2 in the range 40.0–50.0 cm.	1
2(b)(ii)	Percentage uncertainty in L_2 based on absolute uncertainty of 2–10 mm. If repeated readings have been taken, then the uncertainty can be half the range (but not zero) if the working is clearly shown. Correct method of calculation to obtain percentage uncertainty.	1
2(b)(iii)	Correct calculation of L_1 / L_2 .	1
2(b)(iv)	Justification for s.f. in L_1 / L_2 linked to s.f. in L_1 and L_2 .	1
2(c)	Second values of x and L_1 .	1
	Second value of L_2 .	1
	Quality: Second value of $L_2 <$ first value of L_2 .	1
2(d)(i)	Two values of k calculated correctly.	1
2(d)(ii)	Valid comment consistent with calculated values of k , testing against a criterion stated by the candidate.	1
2(e)	Correct calculation of l with a consistent unit.	1

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Question	Answer	Marks
2(f)(i)	A Two readings are not enough to draw a (valid) conclusion (not “not enough for accurate results”, “few readings”). B Difficult to get the lengths the same with reason e.g. judging the strip horizontal, checking the strings vertical, adjusting the setup. C Difficult to determine lengths with a reason e.g. cannot place rule alongside strings, difficult to hold the ruler still, holding ruler with hands, parallax error, difficult to locate centre of bob. D Difficulty with oscillations of rule e.g. have small amplitude, unwanted modes, strings move along during oscillation, stands move, oscillations die away quickly. or Difficult to ensure that x straddles centre of the strip equally. E Difficult to see when both oscillations match their periods. F Releasing bob and rule simultaneously is difficult/uneven force may be applied. <i>1 mark for each point up to a maximum of 4.</i>	4
2(f)(ii)	A Take more readings <u>and</u> plot a graph or take more readings <u>and</u> compare k values (not “repeat readings” on its own). B Detailed use of spirit level/plumb line/set squares. C Mark strings or use a clamped rule/pointer on rule with detail. D Clamp stand to bench/method of attaching string to strip. or Attach/mark scale onto strip or mark centre of strip. E Use longer dimensions (to increase period)/use a video to identify when in phase/use a timer and check individually. F Use a card for a gate for both to release together. <i>1 mark for each point up to a maximum of 4.</i>	4